

Recall game plan:

$$Z_J(z) \equiv \int \frac{d\vec{s}}{(2\pi)^{N/2}} e^{-\frac{1}{2} \vec{s}^T (z\mathbf{I} - \mathbf{J}) \vec{s}}$$

Auxiliary partition function

$$Z(z, n) \equiv \langle Z_J(z)^n \rangle_J$$

average of replicated partition function

$$\nu(z) \equiv \frac{2}{N} \lim_{n \rightarrow 0} \frac{\partial}{\partial n} Z(z, n)$$

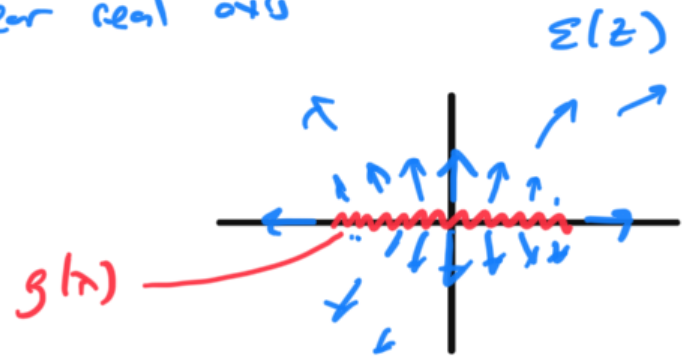
electric potential of a charge density at location z in complex plane where charges are eigenvalues of $\mathbf{J}$ (avg over $\mathbf{J}$)

$$\mathcal{E}(z) = -\partial_z \nu(z)$$

electric field of charge density

$$g(\lambda) = \lim_{\varepsilon \rightarrow 0} \frac{1}{\pi} \text{Im} \mathcal{E}(\lambda - i\varepsilon)$$

charge density from electric field near real axis



Start: average of replicated partition function

$$\begin{aligned} \langle Z_J(z)^n \rangle &= \left\langle \int \prod_{a=1}^n \frac{d\vec{s}^a}{\pi} e^{-\frac{1}{2} \sum_{a=1}^n \vec{s}^{aT} (z\mathbf{I} - \mathbf{J}) \vec{s}^a} \right\rangle_J \\ &= \int \prod_{a=1}^n \frac{d\vec{s}^a}{\pi} e^{-\frac{1}{2} \sum_{a=1}^n \sum_{i=1}^N s_i^a s_i^a} \left\langle e^{+\frac{1}{2} \sum_{a=1}^n \sum_{(i,j)} s_i^a J_{ij} s_j^a} \right\rangle_{P(J_{ij})} \end{aligned}$$

$$P(J_{ij}) = \begin{cases} N(0, \sigma^2/N) & i \neq j \\ N(0, \frac{2\sigma^2}{N}) & i = j \end{cases}$$

Intermediate result:

$$\left\langle e^{\frac{1}{2} \sum_{a=1}^n \sum_{(i,j)} s_i^a J_{ij} s_j^a} \right\rangle_{P(J_{ij})} = e^{N \frac{\sigma^2}{4} \sum_{a,b=1}^n Q_{a,b}^2}$$

replicas decoupled before avg over disorder

replicas coupled after avg over disorder

$$Q_{ab} \equiv \frac{1}{N} \sum_{i=1}^N s_i^a s_i^b$$

replica overlap matrix

$$\Rightarrow \langle Z_J(z)^n \rangle_J = \int \prod_{a=1}^n \frac{d\vec{s}^a}{\pi} e^{N \left\{ -\frac{1}{2} \sum_{a=1}^n Q_{aa} + \frac{\sigma^2}{4} \sum_{a,b=1}^n Q_{ab}^2 \right\}}$$

so

$$\langle Z_J(z)^n \rangle_J = \int \prod_{a=1}^n \pi d\vec{s}^a e^{-N E[Q]}$$

$$E[Q] = \frac{N}{2} \sum_{a=1}^n Q_{aa} - \frac{\sigma^2}{4} \sum_{a,b=1}^n Q_{ab}^2$$

$$\langle Z_J(z)^n \rangle_J = \int \prod_{a \leq b} \pi dQ_{ab} e^{-N E[Q] + N S[Q]}$$

do integral over $\vec{s}^a$
by first integrating
over $\vec{s}^a$ with
a given set of Q_{ab}
then integrating over
 Q_{ab}

$$e^{N S[Q]} \equiv \int \prod_{a=1}^n \pi d\vec{s}^a \delta[\vec{s}^a \cdot \vec{s}^b - N Q_{ab}]$$

$$= \int \prod_{a \leq b} \pi d\hat{Q}_{ab} \int \prod_{a=1}^n \prod_{i=1}^N \pi ds_i^a e^{\sum_{a \leq b} \hat{Q}_{ab} [\sum_{i=1}^N s_i^a s_i^b - N Q_{ab}]}$$

used integral rep of δ -fnctn

$$\delta[s - r] = \int d\hat{q} e^{\hat{q}[s - r]} \quad (\hat{q} \text{ imaginary})$$

$$= \int \prod_{a \leq b} \pi d\hat{Q}_{ab} e^{-N \sum_{a \leq b} \hat{Q}_{ab} Q_{ab}} \int \prod_{a=1}^n \prod_{i=1}^N \pi e^{\sum_{i=1}^N \sum_{a \leq b} \hat{Q}_{ab} s_i^a s_i^b}$$

$$= \int \prod_{a \leq b} \pi d\hat{Q}_{ab} e^{-N \sum_{a \leq b} \hat{Q}_{ab} Q_{ab}} \left[\int \prod_{a=1}^n \pi e^{\sum_{a \leq b} \hat{Q}_{ab} s^a s^b} \right]^N$$

$$= e^{N \Phi[\hat{Q}]}$$

$$\Phi[\hat{Q}] = \ln \int \prod_{a=1}^n \pi e^{\sum_{a \leq b} \hat{Q}_{ab} s^a s^b}$$

Intermediate result:

so

$$e^{N S[Q]} = \int \prod_{a \leq b} \pi d\hat{Q}_{ab} e^{N \left[\sum_{a \leq b} \hat{Q}_{ab} Q_{ab} + \Phi[\hat{Q}] \right]}$$

entropy of
replicated
spins w/
mean overlap

coupling
between dual
variables $\hat{Q}$

(neg) free energy
of a set of replicated
spins with dual variables $\hat{Q}$

given

and original
overlap Q

as coupling constants

Note at large N saddle point approx $\Rightarrow$

$$S[Q] = \text{extr}_{\hat{Q}_{ab}} \left\{ \sum_{a \leq b} \hat{Q}_{ab} Q_{ab} + \Phi[\hat{Q}_{ab}] \right\}$$

entropy $S[Q]$ = Legendre dual of free energy $\Phi[\hat{Q}]$

Note we saw all this before for spin glasses! its exactly the same

$$\langle Z_J(z)^n \rangle = \int \prod_{a \leq b} dQ_{ab} d\hat{Q}_{ab} e^{N \left\{ -E[Q] + \sum_{a \leq b} \hat{Q}_{ab} Q_{ab} + \Phi[\hat{Q}] \right\}}$$

Energy and free energy:



Energy and free energy:

$$E[Q] = \frac{n}{2} \sum_{a=1}^n Q_{aa} - \frac{\sigma^2}{4} \sum_{a \leq b} Q_{ab}^2$$

$$\Phi[\hat{Q}] = \ln \int \prod_{a=1}^n ds^a e^{\sum_{a \leq b} \hat{Q}_{ab} s^a s^b}$$

Now we need to take the $n \rightarrow 0$ limit

try a diagonal replica symmetric ansatz:

$$Q_{ab} = q \delta_{ab}$$

$$Q = \begin{bmatrix} q & & 0 \\ & \ddots & \\ 0 & & q \end{bmatrix}$$

$$\hat{Q}_{ab} = \hat{q} \delta_{ab}$$

$$\hat{Q} = \begin{bmatrix} \hat{q} & & 0 \\ & \ddots & \\ 0 & & \hat{q} \end{bmatrix}$$

$$E[Q] \rightarrow n \left\{ \frac{n}{2} q - \frac{\sigma^2}{4} q^2 \right\}$$

$$\sum_{a \leq b} \hat{Q}_{ab} Q_{ab} \rightarrow n \left\{ \hat{q} q \right\}$$

$$\Phi[\hat{Q}] \rightarrow \ln \int \prod_{a=1}^n ds^a e^{\hat{q} \sum_{a=1}^n s_a^2}$$

$$= \ln \left[\int ds e^{\hat{q} s^2} \right]^n$$

$$= n \ln \left[\int ds e^{\hat{q} s^2} \right]$$

$$= n \ln \sqrt{\frac{\pi}{-\hat{q}}}$$

$$\approx \frac{n}{2} \ln \frac{-\pi}{\hat{q}}$$

$$= -\frac{n}{2} \ln \hat{q} + \text{const}$$

Summary:

Recall:

$$\int ds e^{-\frac{1}{2} a s^2} = \sqrt{\frac{2\pi}{a}}$$

$$\hat{q} = -\frac{1}{2} a$$

$$a = -2\hat{q}$$

$$\int ds e^{\hat{q} s^2} = \sqrt{\frac{2\pi}{-2\hat{q}}} = \sqrt{\frac{\pi}{-\hat{q}}}$$

Summary:

$$\langle Z_J(z)^n \rangle = \int dq d\hat{q} e^{Nn \left[-\frac{z}{2} q + \frac{\sigma^2}{4} q^2 + \hat{q} q - \frac{1}{2} \ln \hat{q} \right]}$$

take $N \rightarrow \infty$: perform saddle point approx to integrals over q and $\hat{q}$

take $n \rightarrow 0$: keep coefficient of linear term in n in the integral:

$$\psi(z) = \frac{2}{N} \lim_{n \rightarrow 0} \frac{\partial}{\partial n} \langle Z_J(z)^n \rangle$$

$$q = \frac{1}{2\hat{q}}$$

$$-\frac{z}{2}$$

$$\psi(z) = \lim_{q, \hat{q}} \left\{ -z q + \frac{\sigma^2}{2} q^2 + 2\hat{q} q - \ln \hat{q} \right\}$$

extremum equations

$$\frac{\partial}{\partial \hat{q}} : 2q = \frac{1}{\hat{q}} \Rightarrow 2\hat{q} = \frac{1}{q}$$

$$\frac{\partial}{\partial q} : -z + \sigma^2 q + 2\hat{q} = 0$$

$$-z + \sigma^2 q + \frac{1}{q} = 0$$

$$\frac{1}{q(z)} = z - \sigma^2 q(z)$$

$$q(z) = \frac{1}{z - \sigma^2 q(z)}$$

self-consistent equation for
replica parameter q as a
function of z

now for the electric field $\varepsilon(z)$

$$\varepsilon(z) = -\partial_z \psi(z)$$

$$-\frac{1}{q(z)^2} \frac{dq(z)}{dz} - \ln \hat{q}(z)$$

Now for the electric field $\varepsilon(z)$

$$\varepsilon(z) = -\partial_z \sqrt{z}$$

$$\sqrt{z} = -zq + \frac{\sigma^2}{2} q(z)^2 + \underbrace{2\hat{q}(z)q(z)}_{\substack{(2\hat{q} = \frac{1}{q}) \\ \text{at saddle point}}} - \underbrace{\ln \hat{q}(z)}_{\ln \hat{q} = \ln \frac{1}{2q}}$$

$$\sqrt{z} = -zq(z) + \frac{\sigma^2}{2} q(z)^2 + \ln q(z) + \text{const}$$

$$\begin{aligned} \varepsilon(z) = -\partial_z \sqrt{z} &= q(z) + z \frac{\partial q}{\partial z} - \sigma^2 q(z) \frac{\partial q}{\partial z} - \frac{1}{q(z)} \frac{\partial q}{\partial z} \\ &= q(z) + \underbrace{\left[z - \sigma^2 q(z) - \frac{1}{q(z)} \right]}_{=0 \text{ at saddle point}} \frac{\partial q}{\partial z} \\ &= q(z) \end{aligned}$$

$$\varepsilon(z) = q^*(z) \quad \text{where } q^*(z) \text{ solves the saddle point equation}$$

$$q^*(z) = \frac{1}{z - \sigma^2 q^*(z)}$$

Magic!: the electric field $\varepsilon(z)$ is the replica symmetric solution $q^*(z)$ to the saddle point equation or resolvent

we will see why $\varepsilon(z)$ satisfies this self-consistent equation using the cavity method

Solving the self-consistent equation for the electric field

$$\varepsilon(z) = \frac{1}{z - \sigma^2 \varepsilon(z)}$$

$$\varepsilon z - \sigma^2 \varepsilon^2 = 1$$

Solving the self consistent equation for the electric field

$$\varepsilon(z) = \frac{1}{z - \sigma^2 \varepsilon(z)}$$

$$\varepsilon z - \sigma^2 \varepsilon^2 = 1$$

$$\sigma^2 \varepsilon^2 - z \varepsilon + 1 = 0$$

$$\varepsilon(z) = \frac{z \pm \sqrt{z^2 - 4\sigma^2}}{2\sigma^2}$$

Good old quadratic formula :)

extracting the charge density from the electric field

$$g(\lambda) = \lim_{\varepsilon \rightarrow 0} \frac{1}{\pi} \operatorname{Im} \varepsilon(\lambda - i\varepsilon)$$

for $z = \lambda - i\varepsilon$ near the real axis $\varepsilon(z)$ has an imaginary part

$$\text{iff } \lambda^2 - 4\sigma^2 < 0$$

$$\text{or } |\lambda| < 2\sigma$$

and the magnitude of the imaginary part is $\sim \frac{\sqrt{4\sigma^2 - \lambda^2}}{2\sigma^2}$

$$\text{so } g(\lambda) = \begin{cases} \frac{\sqrt{4\sigma^2 - \lambda^2}}{2\sigma^2} & \text{for } |\lambda| < 2\sigma \\ 0 & \text{for } |\lambda| > 2\sigma \end{cases}$$

Semicircular law!:

